MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCAQC202 – Reproductive Physiology, Endocrinology and Induced Breeding Techniques

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any 10 out of 10
Each question carries 1 mark

1. Distinguish between Viviparity and Ovoviviparity? [1] [U]

2. What is Terminal Ampoule? Mention its function? [1] [U]

3. What is Vitellogenesis? [1] [U]

4. Explain Synchronous Hermaphroditism with an Example? [1][U]

5. Name the hormones produced by Adenohypophysis? [2][R]

6. What are Releasing Hormones? [2] [U]

7. Mention the hormones produced by the Pericardial organs and their functions? [3][S]

8. Name any two preservatives for pituitary extract in Hypophysation? [4][R]

9. Explain Hypothalamus- Hypophyseal – Gonadal Axis in Fishes? [4] [R]

10. Distinguish between Pelagic and Demersal eggs of Teleosts? [5][U]

[1 x 10=10] Marks

Part B
Short Answer Questions
Answer 5 Questions
Each question carries 6 marks

11. Elaborate upon Sexual Dimorphism in Fishes? [1] [U]

12. Explain the classification of maturity stages in Teleost ovaries, with the help of suitable diagrams. [1][U]

13. Explain Sex Determination in fishes, with suitable examples? [1][U]

14. Explain the structure, Position and functions of Thyroid, Chromaffin tissue, Gastrointestinal glands and Pineal organ in fishes [2][U]

15. Which are the True Endocrine Organs in Crustacea? Mention the structure, location and functions of each [3] [S]

16. Explain the Principle and Procedure of Eyestalk Ablation in Crustacean Hatcheries? [4][R]

17. Which are the different Pharmaceuticals used for Anaesthesia in Fish Handling. Summarise their properties and advantages? [4] [R]

18. Briefly discuss Cryopreservation of Fish gametes? [5] [U]

[6 x 5 = 30] Marks

Part C
Essay Type Questions
Answer any one Question
Each question carries 10 marks

19. Describe the reproductive system in crabs with the help of suitable diagrams and Explain sexual dimorphism in crabs? [1] [U]

20. What is Sex Reversal? Explain the advantages and methods of Sex Reversal in Fishes [4][R]

21. Elaborate upon the environmental control of reproduction in Fishes [4][R]

[10 x 1 = 10] Marks